AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed Systems

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Admission & Exams

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14 /Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: (1.5) Hours**

Instructions:

- Total Marks: 30
- This exam consists of 5 parts:
  - Question 1: 10 marks
  - Question 2: 4 marks
  - Question 3: 4 marks
  - Question 4: 6 marks
  - Question 5: 6 marks
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [10 marks = 5+5]

A) Choose the correct answer from the following (5 marks):

1. **In a von Neumann architecture, what is stored in RAM?**
   a) Only data
   b) Only program instructions
   c) Both program instructions and data
   d) Only the operating system kernel

2. **Which of the following best describes parallel computing?**
   a) Executing instructions one after another on a single CPU
   b) Using multiple compute resources simultaneously to solve a problem
   c) Using a single computer with one processor
   d) A type of network topology for connecting computers

3. **What does the acronym SIMD stand for?**
   a) Single Instruction, Multiple Data
   b) Single Input, Multiple Devices
   c) Multiple Instruction, Single Data
   d) Synchronous Instruction, Multiple Data

4. **In a distributed memory system, how do processors communicate with each other?**
   a) Through shared global memory
   b) Via a message-passing network
   c) Using semaphores and locks
   d) By direct memory access (DMA) to each other's RAM

5. **Which parallel programming model involves all tasks executing the same program but possibly on different data and with different instructions?**
   a) MPMD (Multiple Program, Multiple Data)
   b) SIMD (Single Instruction, Multiple Data)
   c) SPMD (Single Program, Multiple Data)
   d) Data Parallel Model

B) Match the Term in Column A with its correct description in Column B (5 marks):

Column A	Column B
1. Parallel Computing	a. A single computing component with two or more independent actual processing units
2. Serial Computing	b. Using multiple compute resources simultaneously to solve a problem
3. Multi-core Processor	c. A program being executed by a computer
4. Process	d. A sequential flow of control within a process
5. Thread	e. Executing instructions one after another on a single CPU

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Question 2 [4 marks= 2+2]

a) Explain the von Neumann architecture and describe the function of the Control Unit (CU) and RAM.
b) Provide a real-life analogy to illustrate the difference between serial and parallel computing.

Question 3 [4 marks= 2+2]

a) Differentiate between shared memory and distributed memory architectures regarding processor communication.
b) Explain the SPMD (Single Program, Multiple Data) programming model and how it differs from the SIMD model.

Question 4 [6 marks]

Write a Python program that uses a multiprocessing.Manager to create a shared list. Create two child processes: the first appends [1,2,27] and the second appends [28,30]. The parent must wait for both and print the final list.

Question 5 [6 marks]

Write the Docker CLI commands to:
a) Create a bridge network named "my-network".
b) Run a MongoDB container named "mongodb" connected to that network.
c) Run an Nginx container named "my-web" on the same network, mapping host port 8080 to container port 80.

End of questions



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